Plant abstractions for level-flight terminal safety: sufficiency of four families (issue #58)

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Abstract

Issue #58 replaces one plant abstraction by several *families* of parameterised properties $P[\theta](F)$ of the closed-loop frame map F. This note proves, once and independently of the plant, that each family is *sufficient*: if the parameters θ satisfy the family's side conditions (inequalities in θ alone) and F has the property $P[\theta]$, then from every start state the altitude never goes below the floor z. The four families are k-step contraction (Theorem 8), energy barrier (Theorems 14 and 16), slow-fast (Theorem 20) and differential inclusion (Theorem 25). Every proof is in exact real arithmetic and reduces to one k-step invariance lemma (Lemma 1). Each family comes with a satisfying toy map, every hypothesis discharged, and a sinking map that provably fails it. Section 7 connects the families' common conclusion to the city gate's end sets: it supplies exactly the hypotheses `hconf` and `hinv` of `rayEndSet` and `helixEndSet` (`CityGate/EndSets.lean`), with a k-frame version of `CityGate/TermSkeleton.lean`'s per-frame abstraction interface, and lists what is still missing. Membership of the real closed loop in any family is not addressed here.

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1 Setting and conventions

The notation follows `docs/termsound-proof.tex` (branch `eric/informal-proof`).

- X is a set of closed-loop states and $F : X \to X$ the closed-loop frame map (for the city gate, one JSBSim frame $\Delta t = 1/128\,\mathrm{s}$ of `cityGate.fly`; Section 7). $F^0 = \mathrm{id}$ and $F^{n+1} = F \circ F^n$, so $F^{m+n} = F^m \circ F^n$.
- $h : X \to \mathbb{R}$ is the altitude and $z \in \mathbb{R}$ the floor. Issue #58 takes $z = z_{\mathrm{top}} + e$ (934 m plus the airframe margin $e = 7\,\mathrm{m}$). Section 7 shows that $z = z_{\mathrm{top}}$ already suffices for the end sets.
- All quantities are real and all arithmetic is exact. The examples use rational numbers only.
- A *family* is a parameter set Θ, a start set $\mathrm{Start}_\theta \subseteq X$ for each $\theta \in \Theta$, side conditions $\Sigma(\theta)$ (inequalities in θ alone, checkable without the plant), and a property $P[\theta](F)$ of the plant.
- **Sufficiency** of a family is the statement

$$\forall \theta.\ \Sigma(\theta) \wedge P[\theta](F) \implies \forall x \in \mathrm{Start}_\theta.\ \forall n \in \mathbb{N}.\ h(F^n x) \geq z.$$

- The safety claim is written $\mathrm{Safe}(\mathrm{Start}) :\iff \forall x \in \mathrm{Start}.\ \forall n \in \mathbb{N}.\ h(F^n x) \geq z$.

2 The common core

Lemma 1 (k-step invariance)**.** *Let $k \geq 1$ and $I \subseteq X$ with*

(I1) $x \in I \Rightarrow F^k x \in I$*;*

(I2) $x \in I \wedge 0 \leq j < k \Rightarrow h(F^j x) \geq z$*.*

Then $\mathrm{Safe}(I)$*. Moreover the union of images $U = \{F^j y : y \in I,\ 0 \leq j < k\}$ contains I, satisfies $F(U) \subseteq U$, and $h \geq z$ on U.*

Proof. First, $F^{qk} x \in I$ for $x \in I$ and every $q \in \mathbb{N}$, by induction on q: $F^0 x = x$, and $F^{(q+1)k} x = F^k(F^{qk} x) \in I$ by (I1). Given n, divide by k: $n = qk + j$ with $0 \leq j < k$. Then $F^n x = F^j(F^{qk} x)$ with $F^{qk} x \in I$, so $h(F^n x) \geq z$ by (I2).

$I \subseteq U$ with $j = 0$. If $x = F^j y \in U$ then $Fx = F^{j+1} y$, which is in U if $j + 1 < k$; if $j + 1 = k$ then $Fx = F^k y \in I \subseteq U$ by (I1). Finally $h \geq z$ on U is (I2). □

With $k = 1$, (I2) reduces to $h \geq z$ on I and $U = I$. Lemma 1 is `docs/termsound-proof.tex`'s "union of images" lemma with the altitude bound added.

Lemma 2 (between steps from a sink-rate bound)**.** *Let $k \geq 1$, $I \subseteq X$ and $\eta \geq 0$. Suppose that for $x \in I$:*

- $h(F^{j+1} x) \geq h(F^j x) - \eta$ *for* $0 \leq j < k - 1$*;*
- $h(x) \geq z + (k - 1)\eta$*.*

Then (I2) holds.

Proof. By induction on j, $h(F^j x) \geq h(x) - j\eta$ for $0 \leq j \leq k-1$. Hence $h(F^j x) \geq z + (k-1-j)\eta \geq z$. □

In flight terms $\eta = \bar{w}\,\Delta t$, where $\bar{w}$ bounds the sink rate on the intermediate frames. The bound is a claim about $F^j x$ for $x \in I$, not about I alone. This is how `docs/termsound-proof.tex` bounds the altitude between whole seconds (Certificate "level flight, one-second abstraction": $970.2 - 12.9 \geq 957\,\mathrm{m}$).

Lemma 3 (rejection through sufficiency)**.** *Suppose F eventually goes below the floor from every state: $\forall x\ \exists n.\ h(F^n x) < z$. Then for every family proved sufficient and every θ with $\Sigma(\theta)$ and $\mathrm{Start}_\theta \neq \emptyset$, $P[\theta](F)$ is false. In particular this holds for every F with $h(Fx) \leq h(x) - \eta$ for all x and some $\eta > 0$ (a sinking map), since then $h(F^n x) \leq h(x) - n\eta < z$ for $n > (h(x) - z)/\eta$.*

Proof. If $P[\theta](F)$ held, sufficiency would give $h(F^n x) \geq z$ for all n at any $x \in \mathrm{Start}_\theta$, contradicting the hypothesis at that x. □

Each family below also names the clause of $P[\theta]$ that fails, for a sinking map and the example's θ.

3 Family 1: k-step contraction

For a symmetric positive definite $Q \in \mathbb{R}^{m\times m}$ write $\|v\|_Q = \sqrt{v^\top Q v}$. This is a norm: $\|v\|_Q = \|Q^{1/2}v\|_2$.

Definition 4 (k-step contraction family)**.** The parameters are $\theta = (k, m, \pi, A, Q, r, \delta, \mathrm{Fast}, h_0, w)$:

- $k \geq 1$ frames per step;
- an observable $\pi : X \to \mathbb{R}^m$ (for level flight, the errors (e_h, s, e_V) of `docs/termsound-proof.tex`, Section "Terminal safety by abstraction");
- a linear map $A \in \mathbb{R}^{m\times m}$ (the k-frame model);
- $Q \succ 0$ and a radius $r > 0$;
- a residual bound $\delta \geq 0$;
- a predicate $\mathrm{Fast} \subseteq X$ (boxes on the states π leaves out);
- an affine altitude bound $h_0 \in \mathbb{R}$, $w \in \mathbb{R}^m$.

The set is $I_\theta = \{x \in X : \|\pi x\|_Q \leq r,\ x \in \mathrm{Fast}\}$, and $\mathrm{Start}_\theta = I_\theta$.

The **side conditions** $\Sigma_1(\theta)$:

(Σa) $k \geq 1$, $Q = Q^\top \succ 0$, $r > 0$, $\delta \geq 0$;

(Σb) there is $c \geq 0$ with $c^2 Q - A^\top Q A \succeq 0$ and $c\,r + \delta \leq r$;

(Σc) $h_0 \geq z$ and $r^2\, w^\top Q^{-1} w \leq (h_0 - z)^2$.

The **property** $P_1[\theta](F)$: for every $x \in I_\theta$,

(C1) residual: $\|\pi(F^k x) - A\,\pi(x)\|_Q \leq \delta$;

(C2) fast states: $F^k x \in \mathrm{Fast}$;

(C3) between steps: $h(F^j x) \geq z$ for $1 \leq j < k$;

(C4) altitude: $h(x) \geq h_0 + w^\top \pi(x)$.

Remark 5 (checking the side conditions)**.**

- The least c in (Σb) is $\|A\|_Q = \sqrt{\lambda_{\max}(Q^{-1}A^\top Q A)}$. An exact certificate is a rational c and an exact $LDL^\top$ factorisation of $c^2 Q - A^\top Q A$ with $D \geq 0$.

- rung 1's `q_robust_step` (PR #38) asks instead for the budget $(1+t)c^2r^2 + (1+1/t)\delta^2 \le r^2$ with some $t > 0$. The budget implies $c\,r + \delta \le r$, since $2cr\delta \le tc^2r^2 + \delta^2/t$ gives $(cr+\delta)^2 \le r^2$. Conversely, if $cr > 0$ and $\delta > 0$, then $t = \delta/(cr)$ turns the budget into $(cr+\delta)^2 \le r^2$.

- If the residual is measured as a box $|d_i| \le \delta_i$, then (C1) holds with $\delta = \sqrt{K}$, $K = \max\{v^\top Q v : v_i = \pm\delta_i\}$. The function $v \mapsto v^\top Q v$ is convex, so its maximum over the box is at one of the 2^m vertices.

- (Σc) is one scalar inequality. With $w = e_1$ and $\pi_1 = h - h^\star$ it says $h^\star - r\sqrt{(Q^{-1})_{11}} \ge z$: the ellipsoid's lowest altitude is above the floor.

Lemma 6 (robust step in the Q-ellipsoid)**.** *Assume (Σb). If $\|v\|_Q \le r$ and $\|u - Av\|_Q \le \delta$, then $\|u\|_Q \le r$.*

Proof. $\|Av\|_Q^2 = v^\top A^\top Q A v \le c^2\, v^\top Q v$ by $c^2 Q - A^\top Q A \succeq 0$. Since $c \ge 0$, $\|Av\|_Q \le c\,\|v\|_Q \le c\,r$. By the triangle inequality, $\|u\|_Q \le \|Av\|_Q + \|u - Av\|_Q \le c\,r + \delta \le r$. □

Lemma 7 (linear functional on the ellipsoid)**.** *For $Q \succ 0$, $|w^\top v| \le \sqrt{w^\top Q^{-1} w}\;\|v\|_Q$.*

Proof. $w^\top v = (Q^{-1/2}w)^\top (Q^{1/2}v)$, and Cauchy–Schwarz bounds it by $\|Q^{-1/2}w\|_2 \|Q^{1/2}v\|_2$. □

Theorem 8 (sufficiency, k-step contraction)**.** *If $\Sigma_1(\theta)$ and $P_1[\theta](F)$, then* Safe(I_θ)*. Moreover $U_\theta = \{F^j y : y \in I_\theta,\ j < k\}$ is invariant under F and $h \ge z$ on it.*

Proof. We check Lemma 1's hypotheses for $I = I_\theta$.

(I1): let $x \in I_\theta$. Lemma 6 with $v = \pi x$ ($\|v\|_Q \le r$) and $u = \pi(F^k x)$ (C1) gives $\|\pi(F^k x)\|_Q \le r$. (C2) gives $F^k x \in$ Fast. So $F^k x \in I_\theta$.

(I2), $j = 0$: by (C4), Lemma 7 and (Σc),

$$h(x) \;\ge\; h_0 + w^\top \pi x \;\ge\; h_0 - \sqrt{w^\top Q^{-1} w}\, r \;\ge\; z.$$

(The last step uses $h_0 - z \ge 0$ and $r^2 w^\top Q^{-1} w \le (h_0 - z)^2$.) (I2), $1 \le j < k$: this is (C3). □

Remark 9 (what the membership side must show)**.**

- (C1) and (C2) are needed on the *whole* set I_θ, with Fast a stated predicate. A smaller start set $S_0 \subseteq I_\theta$ is safe too, but the residual must still be bounded on all of I_θ.

- `docs/termsound-proof.tex`'s one-second abstraction has this shape: $k = 128$, $\pi = (e_h, s, e_V)$, $A = A_{\text{frame}}^{128}$, $Q = P$, $w = e_1$, $h_0 = h^\star = 1{,}000.22\,\text{m}$. It reports $c_1 = 0.922$ and a measured residual box with $\sqrt{K} = 0.034 \le (1 - c_1)r = 0.152$. Its residual was measured on *settled* states along rollouts, which is a layer in the sense of Family 3 (Section 5), not on a stated Fast.

- (C3) can come from Lemma 2.

Example 10 (non-vacuity)**.** Take $X = \mathbb{R}$, $h(x) = 1000 + x$, $z = 934$ and $F(x) = x/2 + 5$. Let $k = 2$, $m = 1$, $\pi(x) = x$, $A = 1/4$, $Q = 1$, $r = 30$, $\delta = 10$, Fast $= \mathbb{R}$, $h_0 = 1000$, $w = 1$. Then $I_\theta = [-30, 30] \ne \emptyset$.

- (Σa) holds.

- (Σb) with $c = 1/4$: $c^2 Q - A^2 Q = 0 \succeq 0$, and $c\,r + \delta = 7.5 + 10 = 17.5 \le 30$.

- (Σc): $h_0 = 1000 \geq 934$ and $r^2 w^2/Q = 900 \leq 66^2 = 4356$.
- (C1): $F^2 x = x/4 + 15/2$, so the residual is $15/2 \leq 10$ for every x.
- (C2) holds trivially.
- (C3), $j = 1$: $h(Fx) = 1005 + x/2 \geq 990 \geq 934$ for $|x| \leq 30$.
- (C4) holds with equality.

Theorem 8 gives $h(F^n x) \geq 934$ for $|x| \leq 30$. (Directly: $F^n x \to 10$ monotonically.)

Example 11 (rejection)**.** Take the same X, h, z and θ, and the sinking map $G(x) = x - 1$ (one metre per frame). (C1) fails at $x = -30 \in I_\theta$: $G^2(-30) - A \cdot (-30) = -32 + 7.5 = -24.5$, and $24.5 > 10$. Since $h(Gx) = h(x) - 1$, Lemma 3 shows more: no θ with $\Sigma_1(\theta)$ and $I_\theta \neq \emptyset$ has $P_1[\theta](G)$, for any k, π, A or Q.

4 Family 2: energy barrier

Definition 12 (energy-barrier family)**.** The parameters are $\theta = (k, B, b, R)$:

- $k \geq 1$;
- a scalar $B : X \to \mathbb{R}$ (for level flight, the altitude plus weighted energy and climb terms);
- a threshold $b \in \mathbb{R}$;
- a region $R \subseteq X$.

The set is $I_\theta = R \cap \{B \geq b\}$, and $\mathrm{Start}_\theta = I_\theta$. The side condition $\Sigma_2(\theta)$ is $k \geq 1$. The **property** $P_2[\theta](F)$: for every $x \in I_\theta$,

(B1) $F^k x \in R$ and $B(F^k x) \geq b$;

(B2) $h(F^j x) \geq z$ for $1 \leq j < k$;

(B3) $h(x) \geq z$.

Remark 13 (a checkable form of (B3))**.** Suppose $B = h + \varphi$ on R and $\varphi \leq M$ on R; for example $\varphi = w_V(V^2 - V^{\star 2})/(2g) + w_v v_z$ with R bounding V and v_z. Then $b - M \geq z$ implies (B3): on I_θ, $h = B - \varphi \geq b - M \geq z$.

Theorem 14 (sufficiency, energy barrier)**.** *If $\Sigma_2(\theta)$ and $P_2[\theta](F)$, then* Safe(I_θ)*, and $U_\theta = \{F^j y : y \in I_\theta,\ j < k\}$ is F-invariant with $h \geq z$ on it.*

Proof. (B1) is (I1) for $I = I_\theta$, and (B3) with (B2) is (I2). Apply Lemma 1. □

Variant: no drop near the threshold. (B1) asks every state of I_θ to keep $B \geq b$. The variant splits it into a coarse clause far from the threshold and a careful clause in a band of width Δ above it. Far from the threshold, a crude bound on how fast B can fall over k frames is enough. Only the band needs the careful check.

Definition 15 (barrier with margin)**.** The parameters are $\theta' = (k, B, b, R, \varepsilon, \Delta)$. The side conditions $\Sigma_2'(\theta')$ are $k \geq 1$ and $0 \leq \varepsilon \leq \Delta$. The **property** $P_2'[\theta'](F)$: for every $x \in I_\theta$, (B2), (B3), and

(B1a) region: $F^k x \in R$;

(B1b) bounded drop above the band: if $B(x) \geq b + \Delta$ then $B(F^k x) \geq B(x) - \varepsilon$;

(B1c) no drop in the band: if $B(x) < b + \Delta$ then $B(F^k x) \geq B(x)$.

Together, (B1b) and (B1c) say $B(F^k x) \geq B(x) - \varepsilon\,[B(x) \geq b + \Delta]$, where $[\cdot]$ is 1 if the condition holds and 0 otherwise.

Theorem 16 (sufficiency, barrier with margin)**.** *If $\Sigma_2'(\theta')$ and $P_2'[\theta'](F)$, then $P_2[\theta](F)$ for $\theta = (k, B, b, R)$. Hence* Safe(I_θ) *by Theorem 14.*

Proof. Let $x \in I_\theta$. (B1a) gives $F^k x \in R$. If $B(x) \geq b + \Delta$, then by (B1b) and $\varepsilon \leq \Delta$, $B(F^k x) \geq b + \Delta - \varepsilon \geq b$. Otherwise $b \leq B(x) < b + \Delta$, and (B1c) gives $B(F^k x) \geq B(x) \geq b$. So (B1) holds; (B2) and (B3) are assumed. □

(B1c) may be weakened to $B(F^k x) \geq b$ in the band; the proof is unchanged.

Example 17 (non-vacuity)**.** Take $X = \mathbb{R}^2$, $x = (h, u)$ (altitude, climb per frame), $z = 934$ and $F(h, u) = (h + u,\ u/2)$. Let $k = 1$, $B(h, u) = h + 2u$ (the altitude the state will settle at), $b = 1000$ and $R = \{|u| \leq 20\}$. Then $(1000, 0) \in I_\theta$.

- (B1): $B(Fx) = h + u + 2 \cdot u/2 = B(x)$, and $|u/2| \leq 10 \leq 20$.
- (B2) is vacuous ($k = 1$).
- (B3) by Remark 13 with $\varphi = 2u \leq M = 40$ on R: $b - M = 960 \geq 934$.

The altitude itself is not monotone: from $(1040, -20)$ it falls towards 1000. The barrier allows this; a bound on h alone would not.

For the variant, let $\varepsilon = \Delta = 1$ and $F_\sigma(h, u) = (h + u,\ (u - \sigma)/2)$, with $\sigma = 1$ if $B(h, u) \geq b + \Delta$ and $\sigma = 0$ otherwise. This plant gives energy away while it has a margin and holds it in the band. $B(F_\sigma x) = B(x) - \sigma$, so (B1b) holds with drop $1 = \varepsilon$ and (B1c) with no drop. (B1a): $|u - \sigma|/2 \leq 21/2 \leq 20$. (B3) is as before, and Σ_2' holds since $0 \leq 1 \leq 1$.

Example 18 (rejection)**.** $G(h, u) = (h + u,\ (u - 1)/2)$ sinks: $u_n \to -1$, so h eventually falls by almost 1 m per frame. $B(Gx) = B(x) - 1$ everywhere. At $x = (1000, 0) \in I_\theta$, $B(Gx) = 999 < b$, so (B1) fails. In the variant, (B1c) fails at the same x. Lemma 3 applies too, since $h(G^n x) \to -\infty$ from every x.

5 Family 3: slow-fast

Definition 19 (slow-fast family)**.** The parameters are $\theta = (S_0, T, L, \theta_1)$:

- a start set $S_0 \subseteq X$;
- a transient length $T \in \mathbb{N}$ (frames);
- a layer $L \subseteq X$ (fast states near their quasi-steady values);
- Family 1 parameters θ_1 whose fast predicate is the layer, Fast $= L$. Write $I_1 = \{x : \|\pi x\|_Q \leq r\} \cap L$ for Family 1's set.

$\text{Start}_\theta = S_0$. The side conditions $\Sigma_3(\theta)$ are $\Sigma_1(\theta_1)$. The **property** $P_3[\theta](F)$:

(SF1) reach: for each $x \in S_0$ there is $t_x \leq T$ with $F^{t_x} x \in I_1$;

(SF2) transient: $h(F^n x) \geq z$ for $x \in S_0$ and $0 \leq n < T$;

(SF3) slow model on the layer: $P_1[\theta_1](F)$. Its (C1) is the residual on the layer only; its (C2) says the layer is kept from step to step.

Theorem 20 (sufficiency, slow-fast)**.** *If* $\Sigma_3(\theta)$ *and* $P_3[\theta](F)$*, then* $\text{Safe}(S_0)$*.*

Proof. Let $x \in S_0$ and $n \in \mathbb{N}$, and take $t = t_x \leq T$ from (SF1). If $n < t$ then $n < T$, and (SF2) gives $h(F^n x) \geq z$. If $n \geq t$, then $F^n x = F^{n-t}(F^t x)$ with $F^t x \in I_1$. Theorem 8 (applicable by $\Sigma_1(\theta_1)$ and (SF3)) gives $h(F^{n-t}(F^t x)) \geq z$. □

Remark 21.

- The layer need not be invariant frame by frame. Only $F^k(I_1) \subseteq L$ is needed, which is (C2). Entry may happen at any frame up to T, not only at multiples of k: the k-step clock restarts at entry.
- The forward orbit $O(S_0) = \{F^n x : x \in S_0,\ n \in \mathbb{N}\}$ is F-invariant, and by the theorem $h \geq z$ on it. This is the per-frame set Section 7 uses.
- A checkable form of (SF2) is Lemma 2's: $h \geq z + T\eta$ on S_0 and a per-frame drop of at most η along the first T frames.
- In `docs/termsound-proof.tex`'s one-second abstraction, "settled" plays the role of L, and its measured claim "the first 3 s never go below 970 m" is (SF2) with $T = 384$. To use the theorem, L must be a stated predicate, for example boxes around the fast states' quasi-steady values, as that note's "Next" item proposes.

Example 22 (non-vacuity)**.** Take $X = \mathbb{R}^2$, $x = (e, \phi)$ (slow altitude error, fast state), $h = 1000 + e$, $z = 934$ and $F(e, \phi) = (e/2 + \phi,\ \phi/4)$. Let:

- $S_0 = \{|e| \leq 30,\ |\phi| \leq 16\}$, $T = 2$, $L = \{|\phi| \leq 1\}$;
- θ_1: $k = 1$, $\pi(e, \phi) = e$, $A = 1/2$, $Q = 1$, $r = 30$, $\delta = 1$, Fast $= L$, $h_0 = 1000$, $w = 1$.
- $\Sigma_1(\theta_1)$: $c = 1/2$, $cr + \delta = 16 \leq 30$; $1000 - 30 \geq 934$.
- (C1) on I_1: $\pi(Fx) - A\pi(x) = \phi$, and $|\phi| \leq 1$ on L. (C2): $|\phi/4| \leq 1$. (C3) is vacuous. (C4) holds with equality.

- (SF1) with $t_x = 2$: $\phi_2 = \phi/16$, so $|\phi_2| \le 1$. Also $e_1 = e/2 + \phi \in [-31, 31]$ and $e_2 = e_1/2 + \phi/4$, so $|e_2| \le 15.5 + 4 = 19.5 \le 30$.
- (SF2): $h(x) \ge 970$ and $h(Fx) = 1000 + e_1 \ge 969$.

The start set is not inside the layer: $|\phi|$ is up to 16 there. With the same A, Q and r, Family 1 alone would need $\delta \ge 16$, and then $c\,r + \delta = 31 > 30$. The slow-fast family certifies a start set that Family 1, with these parameters, does not.

Example 23 (rejection)**.** $G(e, \phi) = (e - 1,\ \phi/4)$ sinks 1 m per frame. With θ_1 as above, (C1) fails at $(-30, 0) \in I_1$: $\pi(Gx) - A\pi(x) = -31 + 15 = -16$, and $16 > 1$. By Lemma 3, no θ with $\Sigma_3(\theta)$ and $S_0 \ne \emptyset$ has $P_3[\theta](G)$.

6 Family 4: differential inclusion

Definition 24 (differential-inclusion family)**.** The parameters are $\theta = (\psi, \mathcal{G}, \Omega, J, H)$:

- an observable $\psi = (\psi_h, \psi_v, \psi_V) : X \to \mathbb{R}^3$ with $\psi_h = h$ (altitude, climb rate v_z, airspeed V);
- a set-valued envelope $\mathcal{G} : \mathbb{R}^3 \to \mathcal{P}(\mathbb{R}^3)$ of one-frame increments (written F in issue #58; renamed here to keep F for the frame map);
- a region $\Omega \subseteq \mathbb{R}^3$ where the envelope is claimed;
- a candidate invariant set $J \subseteq \mathbb{R}^3$ of the inclusion;
- a predicate $H \subseteq X$ on hidden states ($H = X$ if the envelope holds whatever the hidden states are).

$\mathrm{Start}_\theta = \{x \in X : \psi(x) \in J,\ x \in H\}$.

The **side conditions** $\Sigma_4(\theta)$ involve the inclusion only:

(D1) J is invariant under the inclusion: $y \in J \wedge d \in \mathcal{G}(y) \Rightarrow y + d \in J$;

(D2) J lies above the floor: $y \in J \Rightarrow y_h \ge z$;

(D3) $J \subseteq \Omega$.

The **property** $P_4[\theta](F)$: for every $x \in H$ with $\psi(x) \in \Omega$,

(E1) the plant is in the envelope: $\psi(Fx) - \psi(x) \in \mathcal{G}(\psi(x))$;

(E2) hidden states: $Fx \in H$.

Theorem 25 (sufficiency, differential inclusion)**.** *If* $\Sigma_4(\theta)$ *and* $P_4[\theta](F)$*, then* $\mathrm{Safe}(\mathrm{Start}_\theta)$*, and* Start_θ *is* F*-invariant.*

Proof. Lemma 1 with $k = 1$ and $I = \mathrm{Start}_\theta$. (I1): let $x \in \mathrm{Start}_\theta$, so $\psi(x) \in J \subseteq \Omega$ by (D3). By (E1), $\psi(Fx) = \psi(x) + d$ with $d \in \mathcal{G}(\psi(x))$, so $\psi(Fx) \in J$ by (D1). By (E2), $Fx \in H$. (I2): $h(x) = \psi_h(x) \ge z$ by (D2). □

Remark 26 (checking (D1)–(D3) from envelope tables)**.** Let $J = \prod_i [\alpha_i, \beta_i]$ be a box, covered by finitely many closed cells $C^1, \dots, C^N$ (boxes $\prod_i [a_i^c, b_i^c]$). Let the envelope table give, for each cell, a box $[\ell^c, u^c] \supseteq \mathcal{G}(y)$ for all $y \in C^c$. Then (D1) follows from the $6N$ rational inequalities

$$\alpha_i \le a_i^c + \ell_i^c, \qquad b_i^c + u_i^c \le \beta_i \qquad (i \in \{h, v, V\},\ c = 1..N),$$

since $y + d \in C^c + [\ell^c, u^c] = \prod_i [a_i^c + \ell_i^c,\ b_i^c + u_i^c] \subseteq J$. (D2) is $\alpha_h \ge z$, and (D3) is a box inclusion. State dependence of the increments (for example $\Delta h \ge 0$ near the floor) enters through the cells.

Remark 27. The plant enters only through (E1) and (E2); the reachability over-approximation is (D1)–(D3), a property of θ alone. (E1) must hold for every hidden state allowed by H. The envelope's width therefore has to absorb the unobserved fast states (attitude, rates, α, engine), unless H boxes them and (E2) keeps them boxed.

Example 28 (non-vacuity)**.** Take $X = \mathbb{R}^3$, $\psi = \mathrm{id}$, $y = (h, v, V)$, $z = 934$, $H = X$, and $J = \Omega = [999, 1001] \times [-1, 1] \times [50, 60]$. The envelope is

$$\mathcal{G}(y) = I_h(y_h) \times [-y_v/2 - 1/2,\ -y_v/2 + 1/2] \times [-(y_V - 55)/2 - 1/2,\ -(y_V - 55)/2 + 1/2],$$

with $I_h(y_h) = [0, 1]$ if $y_h < 1000$ and $[-1, 0]$ otherwise.

- (D1), per coordinate. If $999 \le h < 1000$ then $h + d_h \in [999, 1001)$; if $1000 \le h \le 1001$ then $h + d_h \in [999, 1001]$. Next, $v + d_v \in [v/2 - 1/2,\ v/2 + 1/2] \subseteq [-1, 1]$ for $|v| \le 1$. Finally $V + d_V - 55 \in [(V - 55)/2 - 1/2,\ (V - 55)/2 + 1/2] \subseteq [-5, 5]$ for $|V - 55| \le 5$.
- (D2): $999 \ge 934$. (D3): $J = \Omega$.

The plant is $F(h, v, V) = (h + \min(1, \max(-1, 1000 - h)),\ v/2,\ 55 + (V - 55)/2)$.

- (E1): for $h < 1000$, $\Delta h = \min(1, 1000 - h) \in (0, 1]$; for $h \ge 1000$, $\Delta h = \max(-1, 1000 - h) \in [-1, 0]$. Also $\Delta v = -v/2$ and $\Delta V = -(V - 55)/2$, the centres of their intervals.
- (E2) holds trivially.

$\mathrm{Start}_\theta = J \ne \emptyset$.

Example 29 (rejection)**.** $G(h, v, V) = (h - 1,\ v/2,\ 55 + (V - 55)/2)$ sinks. At $y = (999.5, 0, 55) \in J$, $\Delta h = -1 \notin [0, 1]$, so (E1) fails. Lemma 3 excludes every θ with $\Sigma_4(\theta)$ and $\mathrm{Start}_\theta \ne \emptyset$.

7 Connection to the city gate

Instantiation.

- $X = \texttt{Ctl} \times \texttt{Plane}$ and $F = \texttt{cityGate.fly}$. Then $F^n = \texttt{cityGate.flyN}\ n\ (\texttt{flyN}\ (n{+}1)\ x = \texttt{flyN}\ n\ (\texttt{fly}\ x)$, `GateSafety/EndSets.lean`).
- The altitude is the spec's: $h(x) = c_3$ if $\texttt{exactPos}\ x.2 = \texttt{some}\ c$ (the reported CG's altitude above sea level, in metres, read exactly; `CityGate/Geometry.lean`), and $h(x) = z - 1$ if `exactPos` is `none`. So $h(x) \ge z$ implies that `exactPos` is defined.
- The level-off phase is a predicate Lvl on the law's state (terminal law in level-off, held track fixed). It belongs in the family's set: in Fast (Family 1), R (Family 2), L (Family 3) or H (Family 4). The k-step invariance then keeps the phase, and no restricted state type is needed.

Proposition 30 (altitude gives Conf)**.** *Let R be a ray funnel or a helix funnel with release altitude $z^R_{\text{top}} \le z$. If $h(x) \ge z$, then* $\exists c.$ ***exactPos*** $x.2 =$ ***some*** $c \wedge \text{Conf}_R($***toPt*** $c)$.

Proof. `exactPos` is defined, as noted above, and $c_3 = h(x) \ge z \ge z^R_{\text{top}}$. That is the first disjunct of `Ray.Conf` and of `Helix.Conf` (`Spec/EndSets.lean`). □

Proposition 31 (the families give the end sets' `hconf` and `hinv`)**.** *Let* Start_θ *be the start set of any of the four families with* $\Sigma(\theta)$ *and* $P[\theta]($***cityGate.fly***$)$*, and let* $z \ge z^R_{\text{top}}$ *for the funnels used. Define*

$$\text{Dyn}_R(x) \;:\Longleftrightarrow\; z^R_{\text{top}} \le z \;\wedge\; x \in O(\text{Start}_\theta), \qquad O(S) = \{F^n y : y \in S,\ n \in \mathbb{N}\}.$$

Then $\text{Dyn}_R(x) \Rightarrow \exists c.$ ***exactPos*** $x.2 =$ ***some*** $c \wedge \text{Conf}_R($***toPt*** $c)$ *(**hconf**), and* $\text{Dyn}_R(x) \Rightarrow \text{Dyn}_R(Fx)$ *(**hinv**). These are exactly the hypotheses of **rayEndSet** and **helixEndSet** (**CityGate/EndSets.lean**). For Families 1, 2 and 4 the stated set U_θ of Lemma 1 may replace* $O(\text{Start}_\theta)$.

Proof. `hinv`: $F(F^n y) = F^{n+1} y$. `hconf`: for $x = F^n y$ with $y \in \text{Start}_\theta$, the family's theorem gives $h(x) \ge z$; apply Proposition 30. For U_θ, use the invariance and the bound $h \ge z$ in Lemma 1. □

Floor: z_{top} **suffices.** `Ray.Geo` and `Helix.Geo` already require every solid point to lie below $z_{\text{top}} - e$. A reported position at $p_z \ge z_{\text{top}}$ therefore clears the e-box. So $z = z_{\text{top}}$ is enough for the end sets, and issue #58's $z = z_{\text{top}} + e$ is conservative by $e = 7\,\text{m}$.

The k-frame version of the skeleton's interface. `CityGate/TermSkeleton.lean`'s `Abs` is per frame. Its `Dyn` is $(\exists a.\ \pi\, x =$ `some` $a \wedge$ `DynA` $a) \wedge \text{Fast}\ x$, and its obligations are:

- `DynAbstractInv`: one model step is robustly invariant;
- `ResidualBound`: one real frame, `cityGate.fly`, stays within `Near` of the model step;
- `FastBounded`: Fast holds one frame later.

Family 1 with $k = 1$ maps onto it directly: `F` $a = Aa$, `Near` $a\ a' :\Longleftrightarrow \|a' - a\|_Q \le \delta$, `DynA` $= \{\|a\|_Q \le r\}$. Then `DynAbstractInv` is Lemma 6, `ResidualBound` is (C1), `FastBounded` is (C2), and `DynConf` follows from (C4), (Σc) and Proposition 30. For $k > 1$, the measured case ($k = 128$), the per-frame obligations are the wrong shape: the per-frame residual is 49× over budget. The k-frame interface replaces them by:

- $\texttt{ResidualBound}_k$, with `F` now the k-frame model:

$$\forall\, \theta\, x\, a.\ \text{Dyn}\,\theta\, x \to \pi\, x = \texttt{some}\ a \to \exists a'.\ \pi(\texttt{flyN}\ k\ x) = \texttt{some}\ a' \wedge \texttt{Near}\,\theta\,(\texttt{F}\,\theta\, a)\, a';$$

- $\texttt{FastBounded}_k$: $\text{Dyn}\,\theta\, x \to \text{Fast}\,\theta\,(\texttt{flyN}\ k\ x)$;
- $\texttt{Between}_k$: $\text{Dyn}\,\theta\, x \to j < k \to \exists c.$ `exactPos` $(\texttt{flyN}\ j\ x).2 =$ `some` $c \wedge \text{Conf}\,\theta\,(\texttt{toPt}\ c)$ (this is (C3) through Proposition 30; $j = 0$ is `DynConf`);
- `DynAbstractInv` unchanged, for the k-frame model.

The end set's predicate is the union of images $\text{Dyn}^*\,\theta\, x :\Longleftrightarrow \exists j < k.\ \exists y.\ \text{Dyn}\,\theta\, y \wedge x = \texttt{flyN}\ j\ y$. By Lemma 1, Dyn^* is invariant under `cityGate.fly` (`hinv`), and $\texttt{Between}_k$ is its `hconf`. So Dyn^* plugs into `rayEndSet` and `helixEndSet` unchanged. `Ends.Sound` and `InTubes` would take Dyn^* in place of Dyn; `EndTestCertified` then needs end-test states in Dyn^*, and Dyn ($j = 0$) suffices. This change is not made in Lean: the direction of this work changed to informal proofs. The proof is Lemma 1 with `flyN`'s composition law, about 40 lines.

What is still missing for `TermSoundR` on the level-off.

1. **Membership:** $P[\theta]$(`cityGate.fly`) for some family and θ (the other agents of #58). For Family 1, (C1) and (C2) are needed on the whole stated set, not on rollout-settled states. Rollout-settled states call for Family 3 with a stated layer L.

2. **End-test link** (`EndTestCertified`, open in `TermSkeleton.lean`): a passing end test must put the state in Start_θ (or in Dyn^*), including the phase predicate Lvl. The climb through z_{top} into the level set (`docs/termsound-proof.tex`, "The climb") is not covered by any family here.

3. **Same altitude:** the families' h must be `exactPos`'s third coordinate (the reported `hsl` in metres). An abstraction whose π uses another altitude needs a proved relation to it. Under the exact-real assumption the reported altitude is that quantity; the `Float`-to-`Rat` gap belongs to the formal proof.

4. **Release altitude:** $z \geq z^R_{\mathrm{top}}$ for every funnel used. It holds for $z = z_{\mathrm{top}} + e$, and for $z = z_{\mathrm{top}}$.

5. **Airframe:** `AirframeNear 7` stays a hypothesis of PR #29's theorem, as in `docs/termsound-proof.tex` (S4).

8 Summary

Family	$P[\theta](F)$ on the set	Side conditions $\Sigma(\theta)$	Theorem	Examples
k-step contraction	(C1) Q-residual $\leq \delta$ over k frames; (C2) Fast kept; (C3) $h \geq z$ between steps; (C4) $h \geq h_0 + w^\top \pi$	$c^2 Q \succeq A^\top Q A$, $cr + \delta \leq r$; $h_0 \geq z$, $r^2 w^\top Q^{-1} w \leq (h_0 - z)^2$	8	10, 11
energy barrier	(B1) F^k keeps $R \cap \{B \geq b\}$; (B2) between steps; (B3) $h \geq z$ on the set	$k \geq 1$ (and $b - M \geq z$ for (B3) via Remark 13)	14	17, 18
barrier, margin	(B1a) region; (B1b) drop $\leq \varepsilon$ above the band; (B1c) no drop in the band; (B2); (B3)	$0 \leq \varepsilon \leq \Delta$	16	17, 18
slow-fast	(SF1) reach I_1 within T; (SF2) $h \geq z$ for $n < T$; (SF3) Family 1 on the layer	$\Sigma_1(\theta_1)$	20	22, 23
differential inclusion	(E1) $\Delta\psi \in \mathcal{G}(\psi)$ on $\Omega \cap H$; (E2) H kept	(D1) J invariant under $\mathcal{G}$; (D2) J above z; (D3) $J \subseteq \Omega$	25	28, 29

Nothing is assumed beyond the stated hypotheses: every theorem is proved in exact real arithmetic from its side conditions and plant property. Each family's hypotheses are satisfied together by a concrete map, and each family rejects a sinking map, by a named clause and by Lemma 3.